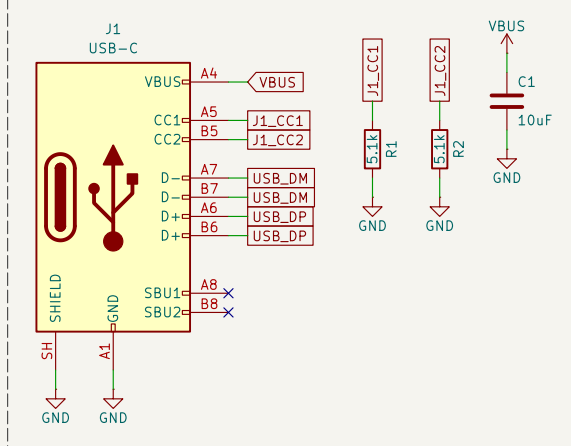
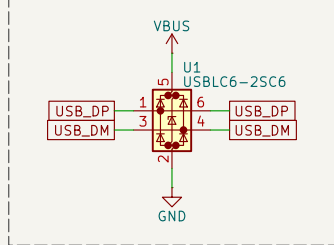


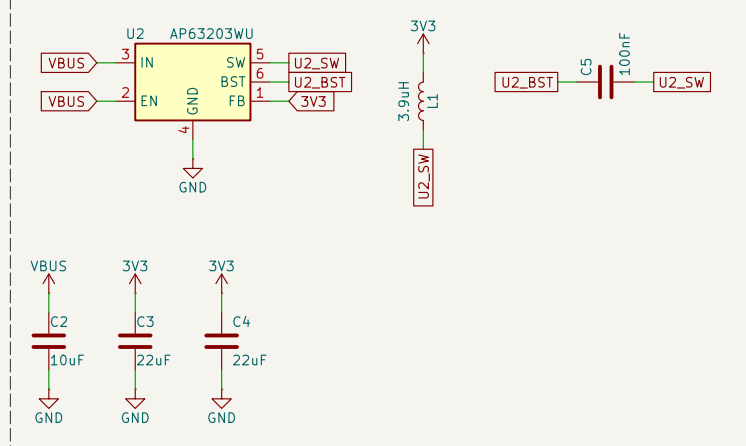
USB-C: 5 V in + USB 2.0 data



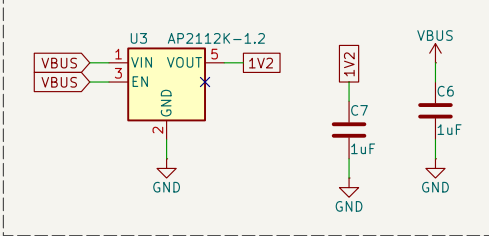
USB ESD clamp



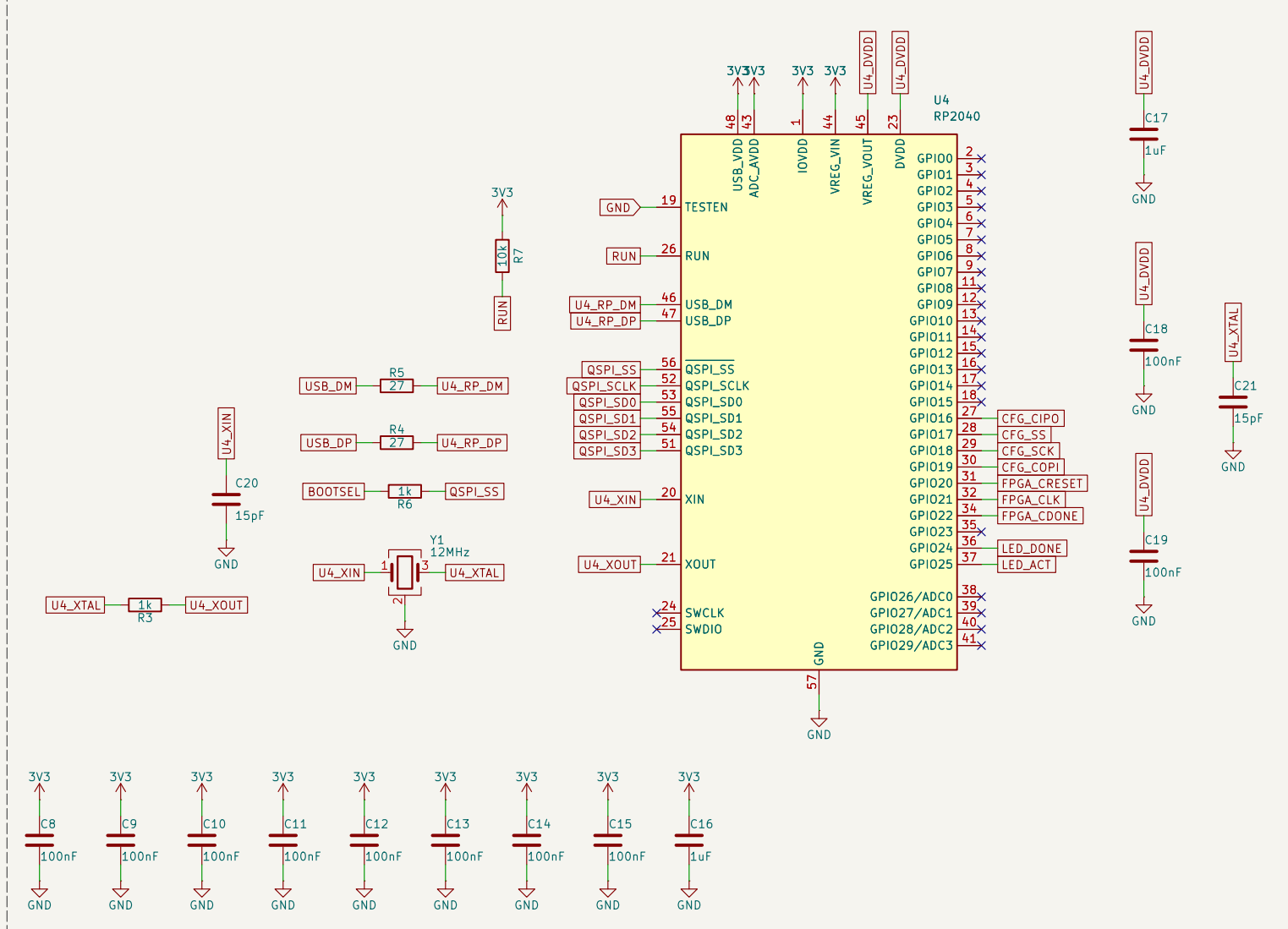
3.3 V buck (AP63203)



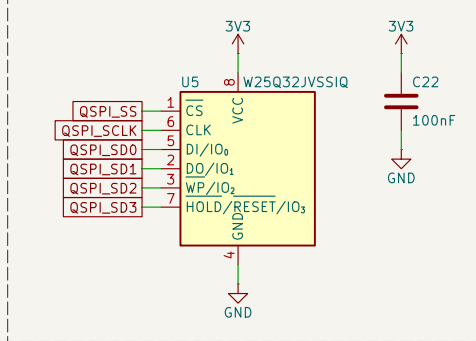
1.2 V core LDO (AP2112K-1.2)



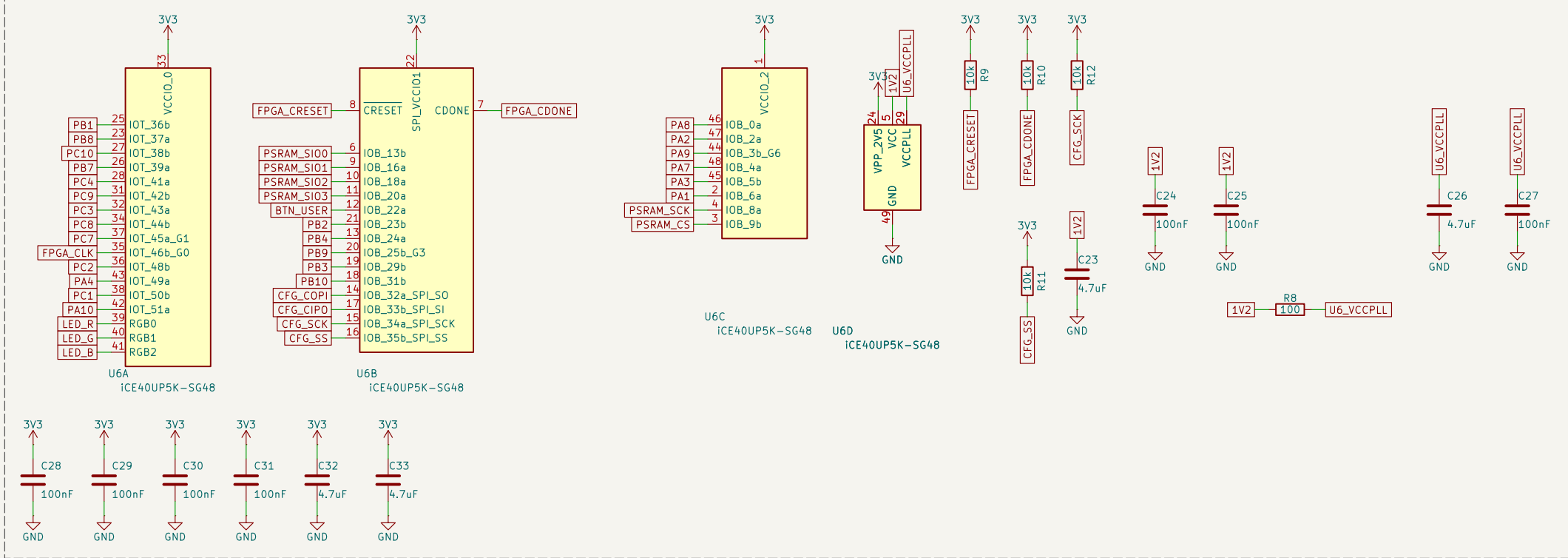
RP2040: USB bridge and FPGA programmer



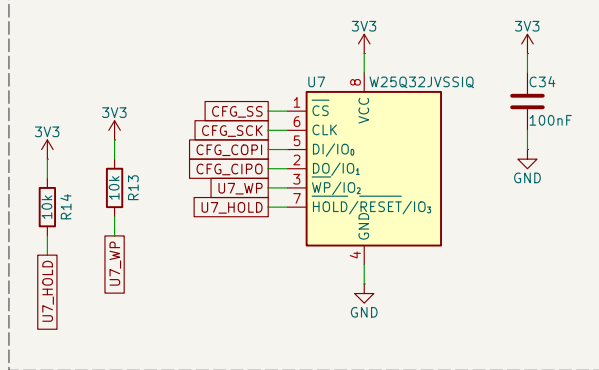
RP2040 QSPI boot flash



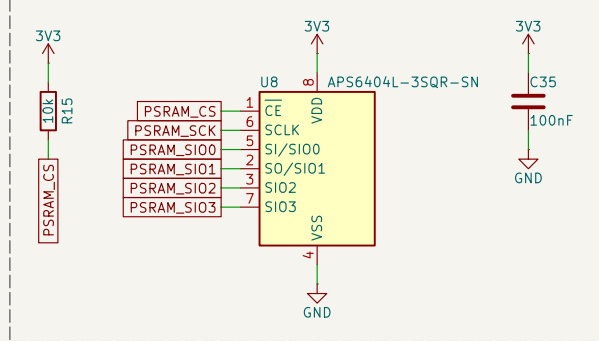
ICE40UP5K-SG48



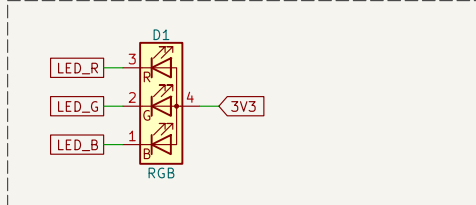
FPGA configuration flash



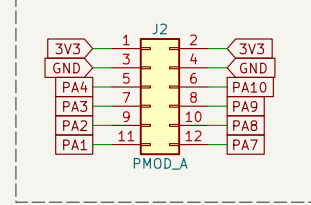
8 MB QSPI PSRAM



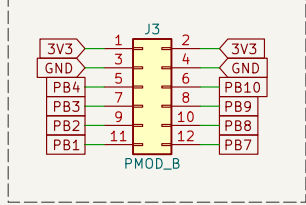
RGB LED on the ICE40 RGB driver



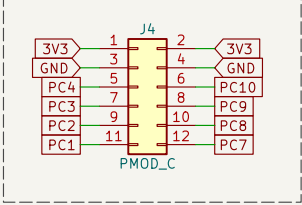
Pmod A



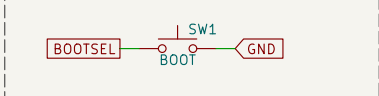
Pmod B



Pmod C



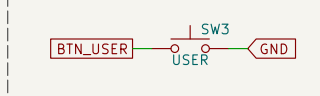
BOOTSEL button (RP2040)



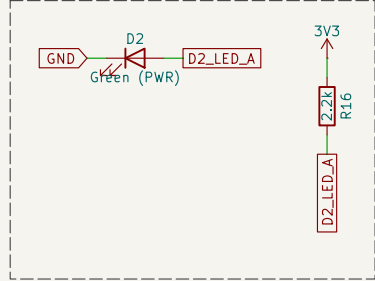
RESET button (RP2040 RUN)



USER button (FPGA)



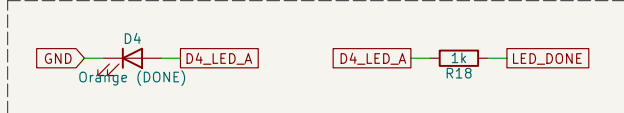
3V3 power LED



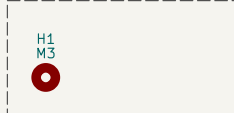
RP2040 activity LED (GPIO25)



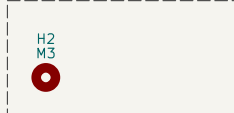
FPGA DONE LED (RP2040 GPIO24)



Mounting hole



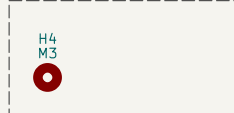
Mounting hole



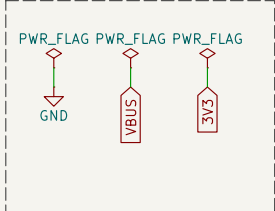
Mounting hole



Mounting hole



Power flags



Generated from blocks -> design.json by heypcb-kicad

HeyPCB method / pure KiCad

Sheet: /

File: ice_core.kicad.sch

Title: Ice Core - ICE40UP5K FPGA + RP2040 dev board, 6 layers

Size: A2 Date: 2026-10-01

KiCad E.D.A. 10.0.6

Rev: A

Id: 1/1